

Personal Opinions on the Basic Questions of AGI

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Introduction

The more you learn about artificial intelligence, the more you realize how far away the finish line actually is. It is vast, it is always changing, and the deeper you go, the more questions come up. I think that confusion is something that will always be there, not just in AI, but in life itself. As humans, we constantly try to figure things out, and sometimes, after all that effort, we realize that not everything has to be fully figured out. Maybe AI is the same way. Maybe it does not have to be perfect. Maybe it just has to be there, helping us live our lives, and we just have to learn how to work with it.

That is the mindset I bring into this paper. It is Spring 2026. I am a Computational Data Science student at Temple University, taking CIS 5590, Artificial General Intelligence, with Professor Pei Wang. Before this class, my understanding of AI was surface level. It is smart software. It helps with tasks. It keeps improving. But once we started going deeper into what intelligence actually means, the difference between narrow AI and general intelligence, and the decades of research behind it, I realized this field is not just technical. It is philosophical. It forces you to think about thinking itself.

Even the definition of AI is not stable. It has changed over time depending on the tools people had, the problems they cared about, and what they expected machines to do. In the 1950s, people like Alan Turing and John McCarthy talked about machines that could think. Since then, the field has split into many directions, each with its own idea of intelligence. Even now, people at the top do not agree on what AGI means. Sam Altman said AGI has become a sloppy term. Jensen Huang has said we already have it. Others say we are still decades away. The fact that there is no clear agreement says a lot about how complex this space is.

Still, all of this can be grounded in four core questions that Professor Pei Wang emphasizes. What is AGI. Can AGI be built. How should we build AGI. Should AGI be built. These questions cut through the noise and give structure to the discussion. This paper is my attempt to

answer them, not as an expert, but as a student who is actively learning and forming my own perspective. Everything here comes from what I have studied in this course, combined with how I think about the material.

Question 1: What is AGI?

Artificial General Intelligence is something that is made to do everything in general. That sounds simple, does it not? But after spending a semester studying this concept, I found out that "general" is not so general. AGI does not just help you with everything in a broad sense. Its capability is beyond what we casually call "general AI." It is not general AI. It is AI that can think generally. That distinction alone makes you think harder, and that is just the start.

Professor Pei Wang defines intelligence as "the ability to adapt to the environment while working with insufficient knowledge and resources." This is the working definition that drives the design of NARS (Non-Axiomatic Reasoning System), a project Professor Wang has developed for over three decades. What makes this definition powerful is what it does not say. It does not say intelligence means getting the right answer. It does not say intelligence means having access to all the data. It says intelligence is the ability to work with what you have, even when what you have is not enough. This is a fundamentally different starting point than what most AI systems are built on today.

IBM defines AGI as "a hypothetical stage in the development of machine learning in which an artificial intelligence system can match or exceed the cognitive abilities of human beings across any task." That is a clean definition, but it is also aspirational. It describes an endpoint without specifying how we would know we got there. The challenge, as IBM also notes, is both philosophical and technological: we need to define what intelligence is before we can build it into a machine, and there is no consensus on that definition.

AGI, at its core, is about a system that learns continuously from experience and keeps updating itself, similar to how the human brain works. Think about a baby. A baby is not born knowing how to walk. It tries, falls, observes, and tries again. Over time, through repetition and experience, it learns. Then it keeps going. It learns how to talk, how to reason, how to handle new situations it has never seen before. That process never really stops. It is always building on

past experience with different levels of confidence. AGI, in theory, should work the same way. It should face new situations, use what it already knows, and adapt without needing to be retrained every time something new appears.

What separates AGI from narrow AI is this ability to transfer and adapt. Most AI systems today are still narrow. They are built for specific tasks, and they do those tasks extremely well. Sometimes even better than humans. But they do not generalize. A language model can write essays, but it cannot learn how to drive. A chess engine can beat top players, but it cannot hold a conversation. As François Chollet explains with the ARC benchmark, these systems mostly follow a memorize, fetch, apply pattern. They perform well on tasks they have effectively seen before, but struggle when something is truly new. What they lack is fluid intelligence, the ability to take what they know and recombine it into something new when facing a completely unfamiliar problem.

So when I say AGI is about thinking generally, I mean it is about building a system that does not just recall patterns. It reasons. It adapts. It handles situations it was never explicitly prepared for. That is what makes it general, and that is what makes it so difficult.

Question 2: Can AGI Be Built?

Yes, I believe AGI can be built. But would it be easy? Absolutely not. If it were easy, every major AI company would have already done it. Claude, ChatGPT, Gemini, all of these systems are remarkable, but none of them are AGI. The capability that AGI demands is beyond everything that currently exists. It is, in a sense, superhuman, or at least, it requires a kind of processing and reasoning that goes far beyond what any single system can do today.

The fundamental challenge is computational. The human brain contains approximately 86 billion neurons, with roughly 100 trillion synaptic connections between them. Each neuron communicates through complex electrical and chemical signals that involve both digital and analog processes. As a 2023 paper published in *Frontiers in Cellular Neuroscience* argues, the brain's computational architecture is fundamentally different from anything we have built in silicon. It is not just a matter of scale. Biological neurons do things that artificial neurons do not: they use gap junctions for analog information processing, they involve glial cells in computation,

and they process information in ways that we still do not fully understand. This is what makes Moravec's paradox so relevant: it is comparatively easy to make computers perform well on intelligence tests or play chess, but extremely difficult to give them the skills of a one year old when it comes to perception and movement.

The energy gap alone is hard to ignore. The human brain runs on about 20 watts, around the same as a dim light bulb. With that, it handles vision, language, movement, memory, creativity, everything at the same time. To get anywhere close to that level of computation with current AI systems, you are talking about energy on the scale of a city. One light bulb versus power for over a million people. That difference makes it clear how far we still have to go.

Open Philanthropy's research on brain computational power suggests that somewhere between 10^{13} and 10^{17} floating point operations per second (FLOP/s) might be sufficient to match the brain's task performance. We have supercomputers that can reach those raw numbers now. But raw computation is not the bottleneck. The bottleneck is architecture, organization, and understanding. We can simulate the speed, but we cannot replicate the structure. Nobody has been able to fully replicate the human brain with all its neurons, synapses, glial cells, and the emergent properties that arise from their interactions. Trying to build a computer that thinks even as well as a human brain, let alone better, remains one of the hardest unsolved problems in science and engineering.

This is also why large language models, despite their impressive performance, are an easier path than true AGI. LLMs are easier to train because they optimize for a specific objective: predicting the next token in a sequence. They do not need to understand the world. They need to be statistically accurate about language patterns. The training cost alone illustrates the scale of the challenge: training GPT-4 reportedly cost over \$100 million, and Google's Gemini Ultra was estimated at \$191 million. These are narrow AI systems. The computational and financial requirements for building something that genuinely thinks across all domains would be orders of magnitude greater.

So yes, I believe AGI is possible. But I believe it will take a very long time, and it will require breakthroughs in areas we have not even fully mapped yet. A September 2025 review of surveys of scientists and industry experts from the past 15 years found that most agreed AGI will occur

before the year 2100, but that is an enormous window. It could be 10 years. It could be 70. The honest answer is that nobody knows for certain.

Question 3: How Should We Build AGI?

If we really have to build this, here is how I think it should be approached. First, let us talk about the obvious: money. AGI development requires massive resources. But money alone will not get us there. We need the right mindset, and the right mindset starts with managing expectations.

The purpose of AGI should be to help us. That is it. Making it like a human, or better than a human, is something close to impossible, and I say that not as a technical limitation but as a philosophical one. Even humans cannot become better humans in any absolute sense. We evolve, we grow, we adapt, but we never reach a final form. Why would we expect a computer to do what we ourselves cannot? Even if a system reaches the capability to match human cognition, humans will have already changed by then. It becomes a never ending loop: we try to build something that matches us, but we are a moving target. AGI development must be grounded in this understanding.

Professor Wang's approach through NARS offers one of the most principled frameworks for thinking about this problem. Rather than trying to replicate human behavior or optimize for benchmarks, NARS is built around a single core principle: intelligence means adaptation under insufficient knowledge and resources. The system is designed to reason with what it has, not with what it ideally should have. It uses non-axiomatic logic, meaning it does not start from fixed assumptions about the world. Instead, it builds knowledge incrementally through experience, revising its beliefs as new evidence comes in. This approach stands in contrast to the dominant paradigm of machine learning, which typically requires massive datasets and predefined objective functions.

I think the right approach to building AGI involves several key principles. Build slowly and deliberately. The rush to claim AGI milestones has led to inflated promises and sloppy definitions. Serious AGI research requires patience. As the literature consistently shows, there are no shortcuts to general intelligence. Take a unified approach. Professor Wang advocates for starting with a core system built on a single technique that captures the fundamental principle of intelligence, rather than stitching together many specialized modules. This creates coherence and

ensures that the system's intelligence is not just a collection of narrow skills wearing a trenchcoat. Prioritize adaptability over performance on benchmarks. Current AI benchmarks reward systems that score well on predetermined tasks. But real intelligence is about handling what you have never seen before. Chollet's ARC-AGI benchmark attempted to measure this kind of adaptability, and the results were telling: GPT-3 scored zero, GPT-4 scored near zero, and even GPT-4o only reached about 5%. These models have enormous knowledge but limited fluid intelligence. Accept that AGI will not be a perfect replica of the human mind. Professor Wang himself has written that "it is possible to build a computer system that follows the same laws of thought and shows similar properties as the human mind, but, since such an AGI will have neither a human body nor human experience, it will not behave exactly like a human." This is an important point. AGI does not mean building a digital person. It means building a system that reasons generally, and that system will inevitably be different from us in fundamental ways.

The path to AGI has to be interdisciplinary. It is not just a computer science problem. It pulls from neuroscience, cognitive psychology, philosophy of mind, linguistics, and ethics. No single field has the full picture, and acting like one does usually leads to dead ends. The people who actually make progress are the ones who pull from different areas, combining formal logic with how real brains actually work.

Question 4: Should AGI Be Built?

I know this has been a long discussion, almost like a philosophy class, talking about human nature more than machine intelligence. But is that not the reason AGI, or AI, or even computers, were made in the first place? For the humans. So if we are wondering whether AGI should be made, my answer is: it depends. And that might sound like a cop out, but hear me out, because it genuinely depends on the goal.

Are we making AGI to actually help us? Or are we making it to become like us? If helping humans is the goal, then I truly believe that building tools to help humanity should never be stopped. Every major technological advancement in history, from the printing press to the internet, has been an attempt to extend human capability. AGI would be the next step in that progression: a tool that can reason, adapt, and assist in ways that no previous technology could.

Medicine, education, scientific research, disaster response, all of these fields would benefit enormously from a system that could genuinely think across domains.

But creating something like a human, when humans do not even fully know or understand who they themselves are? That is a different question entirely. That venture might produce useful things along the way, but in its ultimate aim, it risks becoming an exercise in doing things we do not understand about ourselves. We are trying to build a mind when we have not finished understanding our own. The research community acknowledges this tension. A 2025 study published in Scientific Reports emphasized that AGI development raises ethical issues around "subjectivity, safety, ethical responsibility, controllability, and socialization." The risks include existential threats, poor value alignment, and the reality that current regulatory frameworks are not equipped to handle what AGI could become.

The ethics of AGI are not abstract. They show up in things like jobs, inequality, and who actually holds power. If AGI can do most intellectual work, a lot of human roles disappear. So the real question is not just can we build it, but who benefits when we do. If a small group of companies and investors capture most of the value while millions of people get pushed out, then the system is not helping society. It is just concentrating power. That is something that has to be addressed early, not after everything is already built.

There is also the alignment problem. On paper, it sounds simple to say "make AGI follow human values." In reality, people do not agree on those values. Different cultures think differently. Even across generations, what is considered right can change. So building a system that can handle that responsibly is one of the hardest parts of AGI. It is not just technical. It requires human judgment, different perspectives, and a willingness to move carefully instead of rushing.

My position is pretty straightforward. If AGI is built as a tool to genuinely help people, and if it is developed with ethics, patience, and humility about what we do not know, then it is worth building. But if the main drivers are ego, competition, or profit, then we should slow down and really question what we are doing and who it is actually for.

Conclusion

These four questions, what is AGI, can it be built, how should we build it, and should we build it, do not have clean answers. They change as our understanding changes. A few months ago, my answers would have been a lot simpler. After this semester with Professor Wang, working with NARS, reading, and actually thinking about what intelligence means, my answers are more nuanced, more uncertain, and more real.

AGI is not just a technical challenge. It is a mirror. It makes us look at how we think, how we learn, how we decide what matters, and ask if any of that can actually be turned into a machine. Right now, the answer is partial and imperfect, and honestly very difficult. But the process still matters. Even if we never fully reach human level intelligence, trying to build it teaches us something about the brain, about reasoning, about learning, and about thought itself.

I started by saying that the more you learn about AI, the further the goal feels. I think that is still true. But maybe that distance is the point. Maybe the value is not just in reaching AGI, but in what we learn along the way. And maybe, like in life, not everything needs to be fully figured out. Some things we just learn to work with.

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